معمارية الحاسوب ع
د كاد ثالثة

Processor organization

To understand the organization of the processor, let us consider the requirements placed on the processor, the things that it must do:

- **Fetch instruction:** The processor reads an instruction from memory (register, cache, main memory).
- **Interpret instruction:** The instruction is decoded to determine what action is required.
- **Fetch data:** The execution of an instruction may require reading data from memory or an I/O module.
- **Process data:** The execution of an instruction may require performing some arithmetic or logical operation on data.
- **Write data:** The results of an execution may require writing data to memory or an I/O module.

To do these things,

✓ It should be clear that the processor needs to store some data temporarily.

✓ It must remember the location of the last instruction so that it can know where to get the next instruction.

✓ It needs to store instructions and data temporarily while an instruction is being executed.

In other words, the processor needs a small **internal memory.**

Figure 1 is a simplified view of a processor, indicating its connection to the rest of the system via the **system bus**. The major components of the processor are an **arithmetic and logic unit** (ALU) and a **control unit** (CU). The *ALU* does the actual computation or processing of data. The *control unit* controls the movement of data and instructions into and out of the processor and controls the operation of the ALU. In addition, the figure 1 shows a minimal internal memory, consisting of a set of storage locations, called

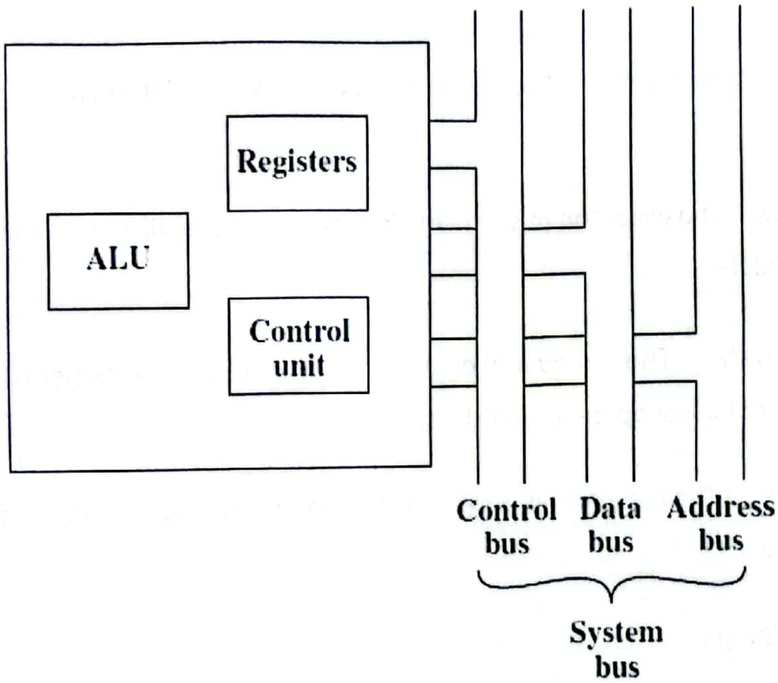

Figure 1: The CPU with the System Bus.

Figure 2 is a slightly more detailed view of the processor. The data transfer and logic control paths are indicated, including an element labeled *internal processor bus*. This element is needed to transfer data between the various registers and the ALU because the ALU in fact operates only on data in the internal processor memory. The figure also shows typical basic elements of the ALU. Note the similarity between the internal structure of the computer as a whole and the internal structure of the processor. In both cases, there is a small collection of major elements (computer: processor, I/O, memory; processor: control unit, ALU, registers) connected by **data paths**.

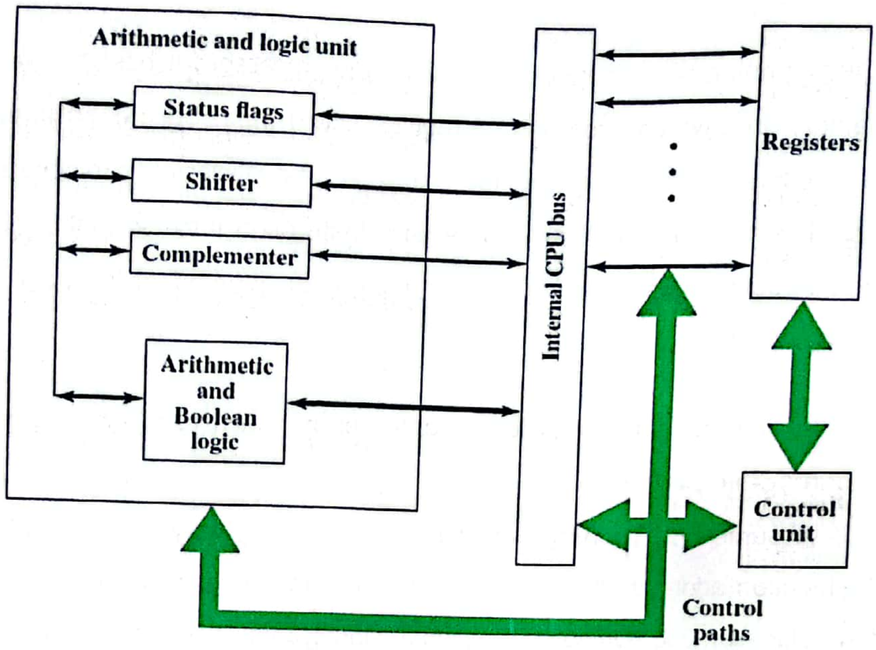

Figure 2: Internal Structure of the CPU.

Instruction Cycle

The basic function performed by a computer is **execution of a program**, which consists of a set of instructions stored in memory. The processor does the actual work by executing instructions specified in the program. In its simplest form, instruction processing consists of two steps:

1. The processor reads (*fetches*) instructions from memory one at a time and
2. executes each instruction.

Program execution consists of repeating the process of instruction fetch and instruction execution. The instruction execution may involve several operations and depends on the nature of the instruction.

The processing required for a single instruction is called an **instruction cycle.** The processor's instruction cycle an instruction cycle includes the following stages:

- **Fetch:** Read the next instruction from memory into the processor.
- **Execute:** Interpret the opcode and perform the indicated operation.
- **Interrupt:** If interrupts are enabled and an interrupt has occurred, save the current process state and service the interrupt.

At the beginning of each instruction cycle, the processor fetches an instruction from memory. In a typical processor, a register called the **program counter** (PC) holds the address of the instruction to be fetched next. Unless told otherwise, the processor always increments the PC after each instruction fetch so that it will fetch the next instruction in sequence (i.e., the instruction located at the next higher memory address).

So, for example, consider a computer in which each instruction occupies one 16-bit word of

Assume that the program counter is set to memory location 300, where the location address refers to a 16-bit word. The processor will next fetch the instruction at location 300. On succeeding instruction cycles, it will fetch instructions from locations 301, 302, 303, and so on. This sequence may be altered, as explained presently. The fetched instruction is loaded into a register in the processor known as the **instruction register** (IR). The instruction contains bits that specify the action the processor is to take. The processor interprets the instruction and performs the required action. In general, these actions fall into four categories:

- Processor-memory: Data may be transferred from processor to memory or from memory to processor.
- Processor-I/O: Data may be transferred to or from a peripheral device by transferring between the processor and an I/O module.
- Data processing: The processor may perform some arithmetic or logic operation on data.
- Control: An instruction may specify that the sequence of execution be altered. For example, the processor may fetch an instruction from location 149, which specifies that the next instruction be from location 182. The processor will remember this fact by setting the program counter to 182. Thus, on the next fetch cycle, the instruction will be fetched from location 182 rather than 150. An instruction's execution may involve a combination of these actions.